Fiscal Year:	FY 2024	**Task Last Updated:**	FY 08/04/2023
PI Name:	Macaulay, Timothy Ph.D.		
Project Title:	Countermeasures for Mitigation of Sensorimotor Decrements Following Head-Down Bed Rest		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058-3711	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
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No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Updates were made to the investigator team, including the confirmation of Dr. Leah Bent, PhD and the addition of Dr. Francisco Valero-Cuevas, PhD. Both have been awarded independent funding (Canadian Space Agency and NASA HRP, respectively) to augment the assessment capabilities of this study. Dr. Bent will provide measures of foot sole skin sensitivity that are based upon her previous measures used in a study on Shuttle astronauts. Dr. Valero-Cuevas will provide a leg dexterity assessment that can be performed in the horizontal position and serve as a transfer task for proprioceptive function. Dr. Eric Rivas, PhD will no longer be involved in this study; however, Dr. Nicole Strock, PhD has joined the team to provide expertise in human performance and exercise science.		
COI Name (Institution):	Bent, Leah Ph.D. (University of Guelph, Guelph, Ontario, Canada) Young, Millennia Ph.D. (NASA Johnson Space Center) Oddsson, Lars Ph.D. (University of Minnesota, Minneapolis) Peters, Brian Ph.D. (KBR/NASA Johnson Space Center) Wood, Scott Ph.D. (NASA Johnson Space Center) Rutkove, Seward (Beth Israel Deaconess Medical Center) Mortreux, Marie (Beth Israel Deaconess Medical Center) Beheshti, Afshin (KBR/NASA Ames Research Center) Bouxsein, Mary (Beth Israel Deaconess Medical Center) Valero-Cuevas, Francisco (Neuromuscular Dynamics LLC) Strock, Nicole (KBR/NASA Johnson Space Center)		

Grant/Contract No.:	Directed Research
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Note: Research aims merged with Grant #80NSSC21K0369 (PI: Rutkove). Astronauts experience postflight disturbances in postural and locomotor control due to sensorimotor adaptations during spaceflight. These alterations may have adverse consequences if a rapid egress is required after landing. Although exercise is effective for mitigating cardiovascular and muscular deconditioning, new approaches are needed to further preserve sensorimotor function, especially during and after exploration missions with less comprehensive exercise capabilities than the current suite on the International Space Station (ISS). We have identified proprioceptive training and electrical muscle stimulation (EMS) as promising in-flight countermeasures. Since prolonged head down bed rest (HDBR) is a spaceflight analog for body unloading and causes postural and locomotor control decrements that parallel those observed after spaceflight, it can be used to accelerate the development of these countermeasures. This study will determine the effects of proprioceptive training and EMS on functional task performance and sensorimotor function following 60 days of 6° HDBR. We will compare four groups of subjects – an EMS arm, a proprioceptive training arm, an exercise plus proprioceptive training arm, and a control arm. The EMS countermeasure will include daily bilateral stimulation of lower extremity muscles (30 minutes per session). Proprioceptive training will be performed three days per week (20 minutes per session) consisting of body-loaded postural tasks in the horizontal position on an air bearing sled. This training aims to keep the proprioceptive and tactile systems conditioned and adaptable to upright balance challenges. Exercise training will mimic current protocols used on the ISS, but treadmill aerobic exercise will be replaced with additional cycling aerobic exercise. Primary outcome measures will include functional tests that are representative of high priority exploration mission tasks and require high demand for dynamic control of postural stability. Additional measures will be used to identify the key physiological factors contributing to countermeasure benefits, including assessments of sensorimotor function, somatosensory feedback, musculoskeletal function, muscle structure using MRI, ultrasound-guided muscle thickness and echointensity, electrical impedance myography, and DXA bone scans. Additionally, serological measurements, neuromuscular biomarker and circulating miRNAs, will be investigated for their potency to be used as biomarkers of muscle degeneration and indicators of treatment response. Associations between these changes in key physiological factors and changes in functional task performance will be explored.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. Exploration missions must be accomplished under new constraints and fewer exercise hardware capabilities. New countermeasures are needed for these missions to replace cumulative sensorimotor stimuli and must be developed in collaboration with other NASA stakeholders. We have designed a countermeasure system to both train and assess proprioceptive function in-flight and are working the Astronaut Strength, Conditioning, and Rehabilitation (ASCR) group and members of the Human Physiology, Performance, Protection, & Operations (H-3PO) Laboratory to define the protocol. The project must be conducted in a spaceflight analog due to testing limitations on the International Space Station (ISS) and the need to determine efficacy in a controlled environment. Therefore, this research effort will require leveraging inhouse expertise including experience with head-down bed rest (HDBR) models to evaluate countermeasures. Outcomes will include functional tests previously established by Johnson Space Center (JSC) laboratories that represent critical mission tasks and have demonstrated performance decrements after both spaceflight and HDBR. It is also essential to track performance and understand the underlying factors that contribute to efficacy in order to translate the countermeasure to spaceflight operations and predict outcomes during exploration missions. The investigators of this research proposal have extensive experience conducting HDBR studies and a strong working relationship with other NASA stakeholders. Proprioceptive training is a novel approach, but a team of sensorimotor and rehabilitation experts to collaboratively define and test the countermeasure increases the likelihood of study success.
Research Impact/Earth Benefits:	The deliverable from this project will be proof-of-concept countermeasure designs for functional task performance with full assessment of efficacy in a spaceflight analog. If the countermeasures are effective, they will be translated for evaluation with the suite of operationally implemented in-flight ISS and exploration countermeasures. This will help mitigate the risk of altered sensorimotor function impacting critical mission tasks. The inclusion of various key physiological measures will also lead to a greater fundamental understanding of how proprioceptive, tactile, and muscular functions contribute to upright functional mobility. These findings may translate to ground-based interventions for various clinical populations that require balance training and fall risk mitigation.
	This past reporting period, we initiated a Science Overview Document with our Research Operations & Integration (ROI) teammates to coordinate study activities across multiple projects/investigators. All activities tied to countermeasures and outcome measures were included. With schedules and time estimates for each session, this document helps facilitate negotiations with the :envihab head-down bedrest (HDBR) facility of the German Aerospace Center (DLR) in Cologne, Germany. Additional justification was provided to support the request for sham electrical muscle stimulation performed with the control group. The project's budget and schedule were rephased to the latest HDBR schedule from ROI. The first HDBR campaign is planned for September 2024. As part of the budget rescope process, assumptions tied to exercise training and testing hardware were refined with our colleagues in the Human Physiology, Performance, Protection & Operations Laboratory. One major piece of exercise hardware used in previous HDBR studies, the Horizontal Squat Device, was transported from NASA Johnson Space Center (JSC) Temporary Storage to JSC Building 21 for reassembly and restoration. After restoring the device to full functionality, standard operating procedures were practiced and recorded for training operators at the :envihab facility. The Horizontal Squat Device was transported back to JSC Temporary Storage until shipment to the :envihab facility next summer. Proprioceptive countermeasure design enhancements continued as part of the Mars Campaign Office (MCO) Crew Health and Performance (CHP) Crew Health Countermeasures (CHC). The primary goals of this work were to analyze

Task Progress:	data from our proprioceptive assessments pilot study, expand the software feedback system's capabilities, assemble a portable version of the countermeasure system that can be used at the :envihab facility, and develop an actuator loading system that mimics what could be used on the International Space Station. The goal of the assessments pilot study was to explore the use of the proprioceptive countermeasure system as a proprioceptive assessment tool; specifically, to determine the association between performance on a supine assessment and upright functional mobility. Seventeen healthy participants (8 males and 9 females, 27.9 ± 8.5 years, 167.2 ± 9.1 cm, 65.5 ± 12.9 kg) completed three supine tasks and one upright functional mobility obstacle course. The supine tasks performed on the proprioceptive countermeasure tilt board included a two-feet dynamic task requiring 3-degree tilts (TB-Dy3), a two-feet dynamic task requiring 9-degree tilts (TB-Dy9), and a single-leg task challenging static control of the tilt board (TB-St). The obstacle course (6.0 m x 4.0 m area) required participants to duck under, step over, weave through, and turn around obstacles. Half of the course included walking on a 10 cm thick medium-density foam to add a proprioceptive challenge. Statistically significant non-parametric correlations were found between completion time on the functional mobility task and TB-Dy3 targets acquired (rho = -0.73), TB-Dy9 targets acquired (rho = -0.53), and TB-St (dominant leg) time on target (rho = -0.50). These data suggest that participants who perform better on the supine tilt board tasks also perform better during the upright functional task. Therefore, performing the supine proprioceptive assessments during HDBR might help predict post-HDBR upright performance in operationally relevant tasks. However, the effects of training and deconditioning on this relationship are unclear. There was a statistically significant difference between the TB-Dy9 performance of participants with prior TB training versus participants with no prior TB experience, but there were no differences between groups in any other tasks. Additional human-in-the-loop pilot testing is planned for next fiscal year, prior to shipping the portable countermeasure system to the :envihab facility next summer. The goals will be to refine our countermeasure training and assessment techniques and integrate protocols across multiple modalities. Ultimately, this work will support operations in the upcoming HDBR study and foster future development of sensorimotor training and assessment techniques for spaceflight.
Bibliography Type:	Description: (Last Updated: 12/23/2025)
Abstracts for Journals and Proceedings	Macaulay TR, Rutkove SB, Peters BT, Oddsson L, Wood SJ, Bent LR, Reschke MF, Rivas E, Young M, Bloomberg JJ, Mortreux M, Bouxsein ML, Beheshti A, Abitante T, and Newman D. "Countermeasures for mitigation of sensorimotor decrements following head-down bed rest." 2023 NASA Human Research Program Investigators' Workshop, "To the Moon: The Next Golden Age of Human Spaceflight", Galveston, TX, February 7-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, "To the Moon: The Next Golden Age of Human Spaceflight", Galveston, TX, February 7-9, 2023. , Feb-2023
Abstracts for Journals and Proceedings	Bellisle RF, Peters BT, Oddsson L, Wood SJ, and Macaulay TR. "Evaluating the relationships between supine proprioception assessments with upright functional mobility and balance tests." 2023 NASA Human Research Program Investigators' Workshop, "To the Moon: The Next Golden Age of Human Spaceflight", Galveston, TX, February 7-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, "To the Moon: The Next Golden Age of Human Spaceflight", Galveston, TX, February 7-9, 2023. , Feb-2023